



Started on	Thursday, 27 August 2026, 10:28 AM
State	Finished
Completed on	Thursday, 27 August 2026, 10:44 AM
Time taken	15 mins 41 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countZeroes(int a[], int low, int high)
4  {
5      if (low > high)
6          return 0;
7
8      int mid = (low + high) / 2;
9
10
11     if (a[mid] == 0)
12     {
13         return (high - mid + 1) + countZeroes(a, low, mid - 1);
14     }
15     else
16     {
17
18         return countZeroes(a, mid + 1, high);
19     }
20 }
21
22 int main()
23 {
24     int m;
25
26     scanf("%d", &m);
27
28     int a[m];
29
30     for (int i = 0; i < m; i++)
31     {
32         scanf("%d", &a[i]);
33     }
34
35     int result = countZeroes(a, 0, m - 1);
36
37     printf("%d", result);
38
39     return 0;
40 }

```

	Input	Expected	Got	
✓	5	2	2	✓
	1			
	1			
	1			
	0			
	0			

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Back to Course



Sanjayvishnu S 251801291-2025-AIDS ▾



Started on	Thursday, 27 August 2026, 10:52 AM
State	Finished
Completed on	Thursday, 27 August 2026, 11:02 AM
Time taken	10 mins 3 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d", &n);
7
8      int nums[50000];
9
10     for (int i = 0; i < n; i++)
11     {
12         scanf("%d", &nums[i]);
13     }
14
15     int candidate = 0;
16     int count = 0;
17
18     for (int i = 0; i < n; i++)
19     {
20         if (count == 0)
21         {
22             candidate = nums[i];
23             count = 1;
24         }
25         else if (nums[i] == candidate)
26         {
27             count++;
28         }
29         else
30         {
31             count--;
32         }
33     }
34

```

```
35     printf("%d", candidate);
36
37     return 0;
38 }
```

	Input	Expected	Got	
✓	3	3	3	✓
	3 2 3			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Back to Course



Sanjayvishnu S 251801291-2025-AIDS ▾



Started on	Thursday, 27 August 2026, 11:03 AM
State	Finished
Completed on	Thursday, 27 August 2026, 11:04 AM
Time taken	1 min 15 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findFloor(int a[], int left, int right, int x)
4  {
5      if (left > right)
6          return -1;
7
8      int mid = (left + right) / 2;
9
10     if (a[mid] == x)
11         return a[mid];
12
13     if (a[mid] < x)
14     {
15         int result = findFloor(a, mid + 1, right, x);
16
17         if (result == -1)
18             return a[mid];
19
20         return result;
21     }
22
23     return findFloor(a, left, mid - 1, x);
24 }
25
26 int main()
27 {
28     int n, x;
29     int a[10000];
30
31     scanf("%d", &n);
32
33     for (int i = 0; i < n; i++)
34     {
35         scanf("%d", &a[i]);
36     }
37
38     scanf("%d", &x);
39
40     int floor = findFloor(a, 0, n - 1, x);
41
42     printf("%d", floor);
43
44     return 0;
45 }
```

	Input	Expected	Got	
✓	6	2	2	✓
	1			
	2			
	8			
	10			
	12			
	19			
	5			
✓	5	85	85	✓
	10			
	22			
	85			
	108			
	129			
	100			
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Back to Course



Started on	Thursday, 27 August 2026, 11:04 AM
State	Finished
Completed on	Thursday, 27 August 2026, 11:05 AM
Time taken	47 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  void findPair(int arr[], int left, int right, int x)
4  {
5      if (left >= right)
6      {
7          printf("No");
8          return;
9      }
10
11     int sum = arr[left] + arr[right];
12
13     if (sum == x)
14     {
15         printf("%d\n", arr[left]);
16         printf("%d", arr[right]);
17         return;
18     }
19
20     if (sum < x)
21     {
22         findPair(arr, left + 1, right, x);
23     }
24     else
25     {
26         findPair(arr, left, right - 1, x);
27     }
28 }
29
30 int main()
31 {
32     int n, x;
33     int arr[10000];
34
35     scanf("%d", &n);
36
37     for (int i = 0; i < n; i++)
38     {
39         scanf("%d", &arr[i]);
40     }
41
42     scanf("%d", &x);
43
44     findPair(arr, 0, n - 1, x);
45
46     return 0;
47 }
```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Back to Course



Started on	Wednesday, 16 September 2026, 7:10 PM
State	Finished
Completed on	Wednesday, 16 September 2026, 7:10 PM
Time taken	42 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5	12 34 67 78 98
67 34 12 98 78	

Answer:

```

4 {
5     if (low < high)
6     {
7         int i = low;
8         int j = high;
9         int pivot = a[low];
10        int temp;
11
12        while (i < j)
13        {
14            while (a[i] <= pivot && i < high)
15                i++;
16
17            while (a[j] > pivot)
18                j--;
19
20            if (i < j)
21            {
22                temp = a[i];
23                a[i] = a[j];
24                a[j] = temp;
25            }
26        }
27
28        temp = a[low];
29        a[low] = a[j];
30        a[j] = temp;
31
32        quickSort(a, low, j - 1);
33        quickSort(a, j + 1, high);
34    }
35 }
36
37 int main()
38 {
39     int n, a[1000];
40
41     scanf("%d", &n);
42
43     for (int i = 0; i < n; i++)
44         scanf("%d", &a[i]);
45
46     quickSort(a, 0, n - 1);
47
48     for (int i = 0; i < n; i++)
49     {
50         if (i > 0)
51             printf(" ");
52         printf("%d", a[i]);
53     }
54
55     return 0;

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)